PINGALL

```
vboxuser@Prajna:~/Desktop$ sudo python3 topology1.py
*** Adding controller
*** Adding switch
*** Adding hosts
*** Adding links
*** Starting network
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...

=== Network Ready ===
h1=10.0.0.1  h2=10.0.0.2  h3=10.0.0.3  h4=10.0.0.4

*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet>
```

```
(ryu-env) vboxuser@Prajna:~/Desktop$ source ~/ryu-env/bin/activate
(ryu-env) vboxuser@Prajna:~/Desktop$ python -m ryu.cmd.manager traffic_classifier.py
loading app traffic_classifier.py
loading app ryu.controller.ofp_handler
instantiating app traffic_classifier.py of TrafficClassifier
Traffic Classification System STARTED
instantiating app ryu.controller.ofp_handler of OFPHandler
Switch connected: dpid=1
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:1 TOTAL:1
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:2 TOTAL:2
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:3 TOTAL:3
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:4 TOTAL:4
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:5 TOTAL:5
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:6 TOTAL:6
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:7 TOTAL:7
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:8 TOTAL:8
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:9 TOTAL:9
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:10 TOTAL:10
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:11 TOTAL:11
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:12 TOTAL:12
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:13 TOTAL:13
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:14 TOTAL:14
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:15 TOTAL:15
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:16 TOTAL:16
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:17 TOTAL:17
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:18 TOTAL:18
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:19 TOTAL:19
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:20 TOTAL:20
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:21 TOTAL:21
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:22 TOTAL:22
[CLASSIFY] ICMP | 6e:ff:ed:53:35:78 -> c2:f2:cc:b0:91:0a | TCP:0 UDP:0 ICMP:23 TOTAL:23
[CLASSIFY] ICMP | c2:f2:cc:b0:91:0a -> 6e:ff:ed:53:35:78 | TCP:0 UDP:0 ICMP:24 TOTAL:24
$
```

ICMP

```
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=9.23 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=13.9 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=8.10 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2007ms
rtt min/avg/max/mdev = 8.095/10.400/13.871/2.498 ms
mininet>
```

```
S[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:25 TOTAL:25
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:26 TOTAL:26
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:27 TOTAL:27
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:28 TOTAL:28
[CLASSIFY] ICMP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:0 UDP:0 ICMP:29 TOTAL:29
[CLASSIFY] ICMP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:0 UDP:0 ICMP:30 TOTAL:30
```

TCP

```
mininet> h2 python3 -m http.server 80 &
[1] 60796
mininet> h1 wget http://10.0.0.2
--2026-04-22 12:43:28--  http://10.0.0.2/
Connecting to 10.0.0.2:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1458 (1.4K) [text/html]
Saving to: ‘index.html.6’

index.html.6        100%[===================>]   1.42K  --.-KB/s    in 0s

2026-04-22 12:43:28 (20.7 MB/s) - ‘index.html.6’ saved [1458/1458]

mininet>
```

```
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:1 UDP:0 ICMP:30 TOTAL:31
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:2 UDP:0 ICMP:30 TOTAL:32
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:3 UDP:0 ICMP:30 TOTAL:33
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:4 UDP:0 ICMP:30 TOTAL:34
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:5 UDP:0 ICMP:30 TOTAL:35
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:6 UDP:0 ICMP:30 TOTAL:36
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:7 UDP:0 ICMP:30 TOTAL:37
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:8 UDP:0 ICMP:30 TOTAL:38
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:9 UDP:0 ICMP:30 TOTAL:39
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:10 UDP:0 ICMP:30 TOTAL:40
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:11 UDP:0 ICMP:30 TOTAL:41
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:12 UDP:0 ICMP:30 TOTAL:42
[CLASSIFY] TCP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:13 UDP:0 ICMP:30 TOTAL:43
[CLASSIFY] TCP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:14 UDP:0 ICMP:30 TOTAL:44
```

UDP

```
mininet> h2 iperf -s -u &
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
10.0.0.1 - - [22/Apr/2026 12:43:28] "GET / HTTP/1.1" 200 -
mininet> h1 iperf -u -c h2 -t 3
------------------------------------------------------------
Client connecting to 10.0.0.2, UDP port 5001
Sending 1470 byte datagrams, IPG target: 11215.21 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 42410 connected with 10.0.0.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-3.0238 sec   389 KBytes  1.05 Mbits/sec
[  1] Sent 272 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-3.0261 sec   389 KBytes  1.05 Mbits/sec   6.062 ms 0/271 (0%)
mininet>
```

```
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:239 ICMP:30 TOTAL:283
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:240 ICMP:30 TOTAL:284
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:241 ICMP:30 TOTAL:285
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:242 ICMP:30 TOTAL:286
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:243 ICMP:30 TOTAL:287
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:244 ICMP:30 TOTAL:288
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:245 ICMP:30 TOTAL:289
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:246 ICMP:30 TOTAL:290
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:247 ICMP:30 TOTAL:291
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:248 ICMP:30 TOTAL:292
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:249 ICMP:30 TOTAL:293
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:250 ICMP:30 TOTAL:294
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:251 ICMP:30 TOTAL:295
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:252 ICMP:30 TOTAL:296
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:253 ICMP:30 TOTAL:297
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:254 ICMP:30 TOTAL:298
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:255 ICMP:30 TOTAL:299
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:256 ICMP:30 TOTAL:300
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:257 ICMP:30 TOTAL:301
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:258 ICMP:30 TOTAL:302
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:259 ICMP:30 TOTAL:303
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:260 ICMP:30 TOTAL:304
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:261 ICMP:30 TOTAL:305
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:262 ICMP:30 TOTAL:306
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:263 ICMP:30 TOTAL:307
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:264 ICMP:30 TOTAL:308
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:265 ICMP:30 TOTAL:309
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:266 ICMP:30 TOTAL:310
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:267 ICMP:30 TOTAL:311
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:268 ICMP:30 TOTAL:312
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:269 ICMP:30 TOTAL:313
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:270 ICMP:30 TOTAL:314
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:271 ICMP:30 TOTAL:315
[CLASSIFY] UDP | 36:1f:84:dc:fa:cd -> d6:96:ff:d3:ec:96 | TCP:14 UDP:272 ICMP:30 TOTAL:316
[CLASSIFY] UDP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:14 UDP:273 ICMP:30 TOTAL:317
[CLASSIFY] UDP | d6:96:ff:d3:ec:96 -> 36:1f:84:dc:fa:cd | TCP:14 UDP:274 ICMP:30 TOTAL:318
```

MULTIPLE TRAFFIC

```
mininet> h2 ping -c 2 h3
------------------------------------------------------------
Server listening on UDP port 5001
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 5001 connected with 10.0.0.1 port 42410
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-3.0261 sec   389 KBytes  1.05 Mbits/sec   6.063 ms 0/271 (0%)
[  3] WARNING: ack of last datagram failed.
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=16.0 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=6.24 ms

--- 10.0.0.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 6.235/11.136/16.038/4.901 ms
mininet> h3 ping -c 2 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=5.25 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=25.0 ms

--- 10.0.0.4 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1010ms
rtt min/avg/max/mdev = 5.249/15.121/24.994/9.872 ms
mininet>
```